

DECLARATION OF CO-AUTHORSHIP

The declaration is for PhD students and must be completed for each conjointly authored article.

Please note that if a manuscript or published paper has eleven or less authors, all authors must sign a declaration of co-authorship. If it has twelve or more authors, only the PhD student, the corresponding author(s), the senior author, and the principal supervisor need to sign the declaration of co-authorship.

1. Declaration by	
Name of PhD student	Neil Alexandre Scheidwasser
E-mail	neil.clow@sund.ku.dk
Name of principal supervisor	Samir Bhatt
Title of the PhD thesis	Pandemic preparedness in a vector: new representations of data from phylogenetic trees, protein structures, and animal tracking to aid preparedness

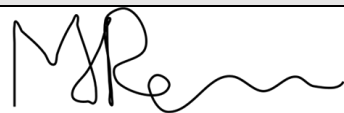
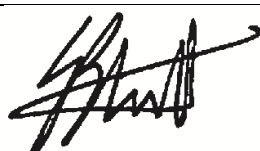
2. The declaration applies to the following article	
Title of article	Phylo2Vec: A Vector Representation for Binary Trees
Article status	
Published ☒ X Date: 27.06.2024	Accepted for publication ☐ Date:
Manuscript submitted ☐ Date:	Manuscript not submitted ☐
If the article is published or accepted for publication, please state the name of journal, year, volume, page and DOI (if you have the information).	<i>Systematic Biology</i> , Volume 74, Issue 2, March 2025, Pages 250-266, https://doi.org/10.1093/sysbio/syae030

3. The PhD student's contribution to the article (please use the scale A-F as benchmark)	
Benchmark scale of the PhD-student's contribution to the article	
A. Has essentially done all the work (> 90 %) B. Has done most of the work (60-90 %) C. Has contributed considerably (30-60 %) D. Has contributed (10-30 %) E. No or little contribution (<10 %) F. Not relevant	A, B, C, D, E, F
1. Formulation/identification of the scientific problem	C
2. Development of the key methods	B
3. Planning of the experiments and methodology design and development	B
4. Conducting the experimental work/clinical studies/data collection/obtaining access to data	A

3. The PhD student's contribution to the article (please use the scale A-F as benchmark) <u>Benchmark scale of the PhD-student's contribution to the article</u> A. Has essentially done all the work (> 90 %) B. Has done most of the work (60-90 %) C. Has contributed considerably (30-60 %) D. Has contributed (10-30 %) E. No or little contribution (<10 %) F. Not relevant		A, B, C, D, E, F
5. Conducting the analysis of data		
6. Interpretation of the results		
7. Writing of the first draft of the manuscript		
8. Finalisation of the manuscript and submission		
Provide a short description of the PhD student's specific contribution to the article. ⁱ Co-first author, corresponding author. I was responsible for developing methods to explain and simplify how phylo2vec works. I also implemented the final software release and conducted research matters pertaining to model evaluation, data visualisation, and complexity analysis. Also corresponded with reviewers, editors, and users of the software.		

4. Material from another thesis / dissertationⁱⁱ	
Does the article contain work which has also formed part of another thesis, e.g. master's thesis, PhD thesis or doctoral dissertation (the PhD student's or another person's)?	Yes: X No: ☐
If yes, please state name of the author and title of thesis / dissertation.	Penn, M. (2024). <i>Mathematical and Statistical Methods in Epidemiology</i> . University of Oxford (United Kingdom).
If the article is part of another author's academic degree, please describe the PhD student's and the author's contributions to the article so that the individual contributions are clearly distinguishable from one another.	MP had the original idea for phylo2vec and was responsible for its mathematical formulation and proofs. I contributed by developing simplifications and explanations of the original formulation of phylo2vec. I also implemented the final software release and conducted research matters pertaining to model evaluation, data visualisation, and complexity analysis.

5. Signatures of the authors incl. the PhD student				
	Date	Name	Title	Signature
1.	26.08.2025	Neil Scheidwasser	MSc	

5. Signatures of the authors incl. the PhD student				
2.	26.08.2025	Matthew J Penn	PhD	
3.	26.08.2025	Mark Poulsen Khurana	MD	
4.	27.08.2025	David A Duchêne	Professor	
5.	26.08.2025	Christl A Donnelly	Professor	
6.	26.08.2025	Samir Bhatt	Professor	

6. Signature of the principal supervisor
I solemnly declare that the information provided in this declaration is accurate to the best of my knowledge. Date: 26.08.2025  Principal supervisor:

7. Signature of the PhD student
I solemnly declare that the information provided in this declaration is accurate to the best of my knowledge. Date: 26.08.2025  PhD student:

Please learn more about responsible conduct of research on the [Faculty of Health and Medical Sciences' website](#).

ⁱ This can be supplemented with an additional letter if needed.

ⁱⁱ Please see Ministerial Order on the PhD Programme at the Universities and Certain Higher Artistic Educational Institutions (PhD Order) § 12 (4):

"Any articles included in the thesis may be written in cooperation with others, provided that each of the co-authors submits a written declaration stating the PhD student's or the author's contribution to the work."